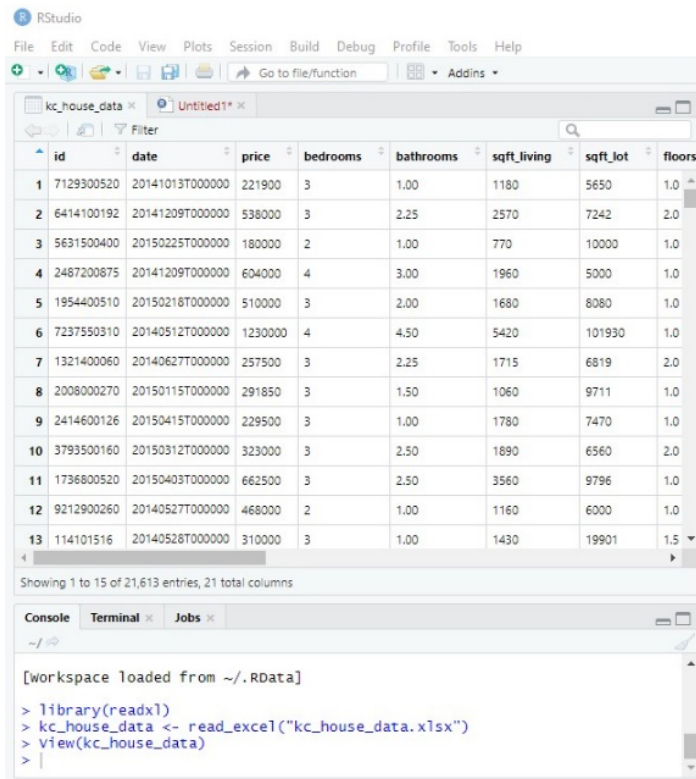


Week 7 Sean Graham

1.

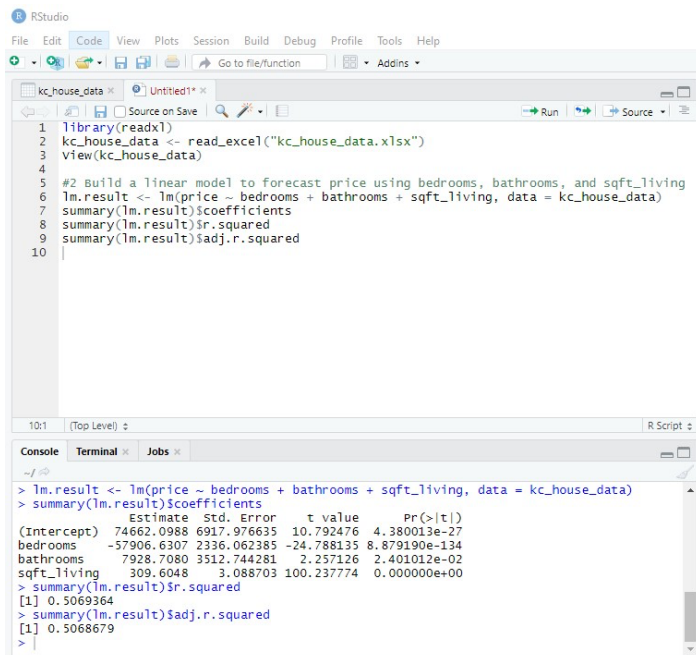


RStudio interface showing a data frame named `kc_house_data` with 13 rows and 8 columns. The columns are `id`, `date`, `price`, `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, and `floors`. The console shows the following code:

```
[workspace loaded from ~/.RData]
> library(readxl)
> kc_house_data <- read_excel("kc_house_data.xlsx")
> view(kc_house_data)
> |
```

2. A. $\text{price} = 74662.0988 + -57906.6307 * \text{bedrooms} + 7928.7080 * \text{bathrooms} + 309.6048 * \text{sqft_living}$

B. Both R square and adjusted R square are about 0.51, so the model isn't very good.



RStudio interface showing the R code to build a linear model and the resulting summary output in the console.

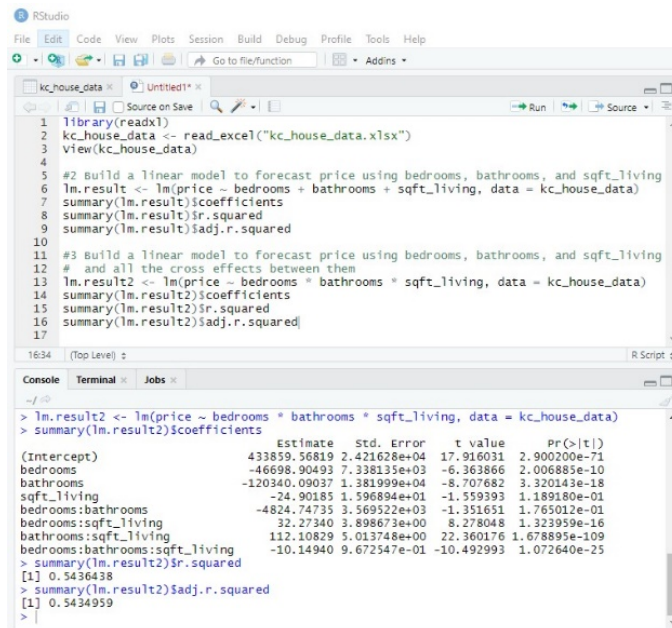
```
1 library(readxl)
2 kc_house_data <- read_excel("kc_house_data.xlsx")
3 view(kc_house_data)
4
5 #2 build a linear model to forecast price using bedrooms, bathrooms, and sqft_living
6 lm.result <- lm(price ~ bedrooms + bathrooms + sqft_living, data = kc_house_data)
7 summary(lm.result)$coefficients
8 summary(lm.result)$r.squared
9 summary(lm.result)$adj.r.squared
10 |
```

Console output:

```
> lm.result <- lm(price ~ bedrooms + bathrooms + sqft_living, data = kc_house_data)
> summary(lm.result)$coefficients
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 74662.0988  6917.976635  10.792476  4.380013e-27
bedrooms    -57906.6307  2336.062385  -24.788135  8.879190e-134
bathrooms     7928.7080  3512.744281   2.257126  2.401012e-02
sqft_living   309.6048    3.088703  100.237774  0.000000e+00
> summary(lm.result)$r.squared
[1] 0.5069364
> summary(lm.result)$adj.r.squared
[1] 0.5068679
> |
```

3. A. $\text{price} = 433859.56819 + -46698.90493 * \text{bedrooms} + -120340.09037 * \text{bathrooms} + -24.90185 * \text{sqft_living} + -4824.74735 * \text{bedrooms} * \text{bathrooms} + 32.27340 * \text{bedrooms} * \text{sqft_living} + 112.10829 * \text{bathrooms} * \text{sqft_living} + -10.14940 * \text{bedrooms} * \text{bathrooms} * \text{sqft_living}$

B. The adjusted R square is about 0.54, so while it's slightly better than the model in question 2, it still isn't very good.



```

1 library(readxl)
2 kc_house_data <- read_excel("kc_house_data.xlsx")
3 view(kc_house_data)
4
5 #2 Build a linear model to forecast price using bedrooms, bathrooms, and sqft_living
6 lm.result <- lm(price ~ bedrooms + bathrooms + sqft_living, data = kc_house_data)
7 summary(lm.result)$coefficients
8 summary(lm.result)$r.squared
9 summary(lm.result)$adj.r.squared
10
11 #3 Build a linear model to forecast price using bedrooms, bathrooms, and sqft_living
12 # and all the cross effects between them
13 lm.result2 <- lm(price ~ bedrooms * bathrooms * sqft_living, data = kc_house_data)
14 summary(lm.result2)$coefficients
15 summary(lm.result2)$r.squared
16 summary(lm.result2)$adj.r.squared
17
1834 (Top Level)
R Script

```

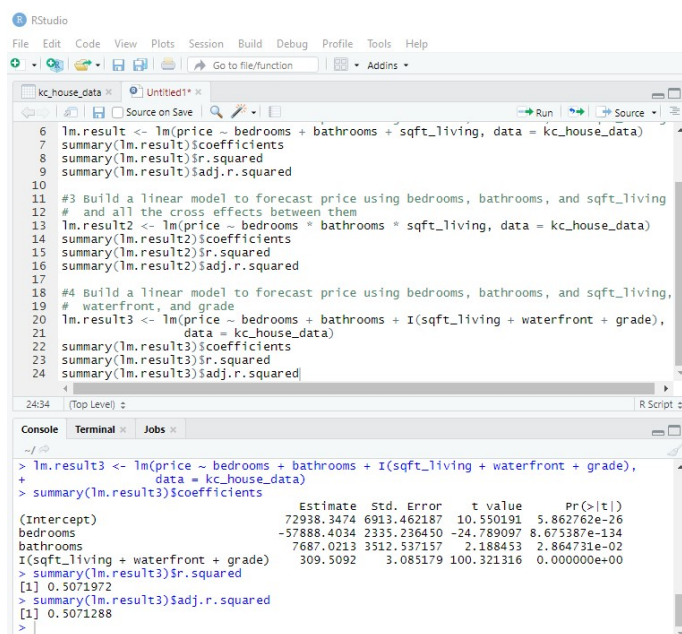
```

> lm.result2 <- lm(price ~ bedrooms * bathrooms * sqft_living, data = kc_house_data)
> summary(lm.result2)$coefficients
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  433859.56819  2.421628e+04  17.916031  2.900200e-71
bedrooms      -46698.90493  7.338135e+03   -6.363866  2.006885e-10
bathrooms     -120340.09037  1.381999e+04   -8.707682  3.320143e-18
sqft_living    -24.90185  1.596894e+01   -1.559393  1.189180e-01
bedrooms:bathrooms -4824.74735  3.569522e+03   -1.351651  1.765012e-01
bedrooms:sqft_living  32.27340  3.898673e+00    8.278048  1.323959e-16
bathrooms:sqft_living 112.10829  5.013748e+00   22.360176  1.678895e-109
bedrooms:bathrooms:sqft_living -10.14940  9.672547e-01  -10.492993  1.072640e-25
> summary(lm.result2)$r.squared
[1] 0.5436438
> summary(lm.result2)$adj.r.squared
[1] 0.5434959
>

```

4. A. # $\text{price} = 72938.3474 + -57888.4034 * \text{bedrooms} + 7687.0213 * \text{bathrooms} + 309.5092 * (\text{sqft_living} + \text{waterfront} + \text{grade})$

B. The adjusted R square is about 0.51, so it's slightly worse than the model in question 3 and it still isn't very good.



```

6 lm.result <- lm(price ~ bedrooms + bathrooms + sqft_living, data = kc_house_data)
7 summary(lm.result)$coefficients
8 summary(lm.result)$r.squared
9 summary(lm.result)$adj.r.squared
10
11 #3 Build a linear model to forecast price using bedrooms, bathrooms, and sqft_living
12 # and all the cross effects between them
13 lm.result2 <- lm(price ~ bedrooms * bathrooms * sqft_living, data = kc_house_data)
14 summary(lm.result2)$coefficients
15 summary(lm.result2)$r.squared
16 summary(lm.result2)$adj.r.squared
17
18 #4 Build a linear model to forecast price using bedrooms, bathrooms, and sqft_living,
19 # waterfront, and grade
20 lm.result3 <- lm(price ~ bedrooms + bathrooms + I(sqft_living + waterfront + grade),
21 data = kc_house_data)
22 summary(lm.result3)$coefficients
23 summary(lm.result3)$r.squared
24 summary(lm.result3)$adj.r.squared
25
2434 (Top Level)
R Script

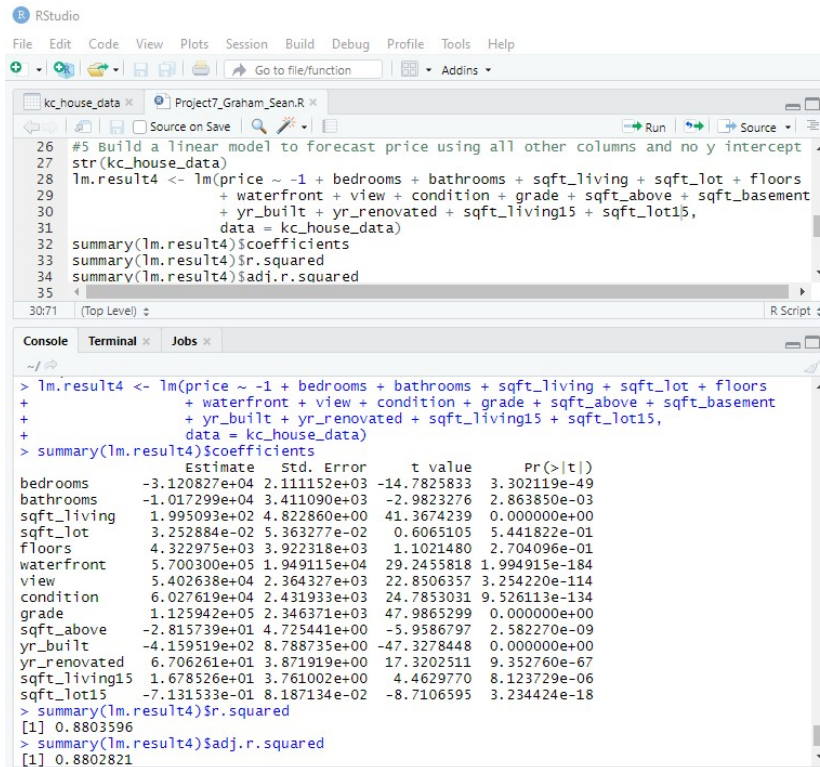
```

```

> lm.result3 <- lm(price ~ bedrooms + bathrooms + I(sqft_living + waterfront + grade),
+ data = kc_house_data)
> summary(lm.result3)$coefficients
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  72938.3474  6913.462187   10.550191  5.862762e-26
bedrooms     -57888.4034  2335.236450   -24.789097  8.675387e-134
bathrooms      7687.0213  3512.537157    2.188453  2.864731e-02
I(sqft_living + waterfront + grade)  309.5092  3.085179  100.321316  0.000000e+00
> summary(lm.result3)$r.squared
[1] 0.5071972
> summary(lm.result3)$adj.r.squared
[1] 0.5071288
>

```

5. The model in question 5 is the best. The adjusted R is about 0.88, which is by far the closest to 1 of all the models thus far.

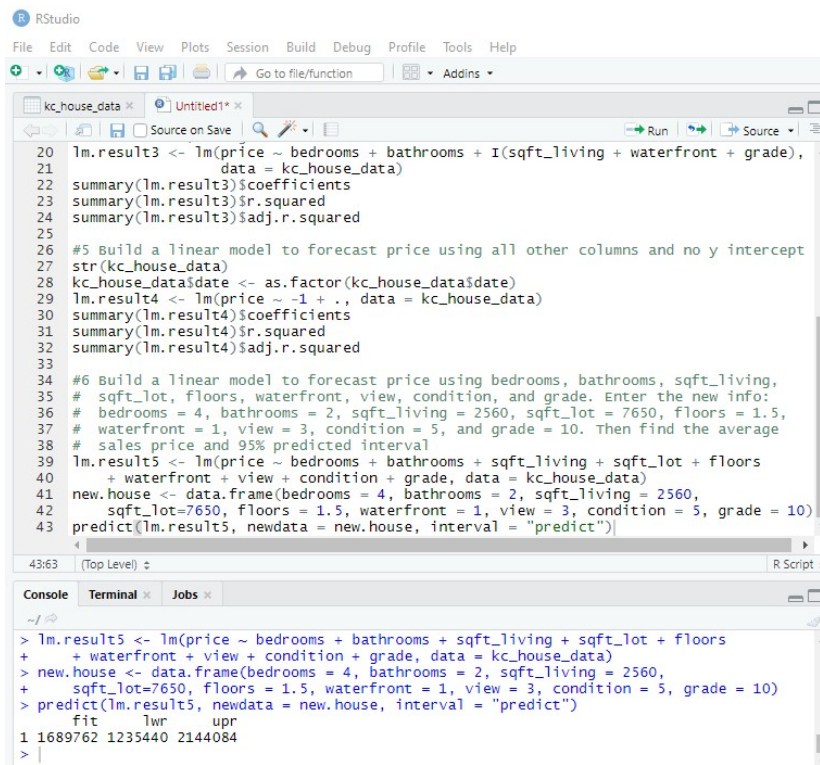


```

26 #5 Build a linear model to forecast price using all other columns and no y intercept
27 str(kc_house_data)
28 lm.result4 <- lm(price ~ -1 + bedrooms + bathrooms + sqft_living + sqft_lot + floors
29 + waterfront + view + condition + grade + sqft_above + sqft_basement
30 + yr_built + yr_renovated + sqft_living15 + sqft_lot15,
31 data = kc_house_data)
32 summary(lm.result4)$coefficients
33 summary(lm.result4)$r.squared
34 summary(lm.result4)$adj.r.squared
35
~/
> lm.result4 <- lm(price ~ -1 + bedrooms + bathrooms + sqft_living + sqft_lot + floors
+ + waterfront + view + condition + grade + sqft_above + sqft_basement
+ + yr_built + yr_renovated + sqft_living15 + sqft_lot15,
+ data = kc_house_data)
> summary(lm.result4)$coefficients
              Estimate Std. Error t value Pr(>|t|)
bedrooms    -3.120827e+04  2.111152e+03 -14.7825833  3.302119e-49
bathrooms   -1.017299e+04  3.411090e+03  -2.9823276  2.863850e-03
sqft_living  1.995093e+02  4.822860e+00  41.3674239  0.000000e+00
sqft_lot     3.252884e-02  5.363277e-02   0.6065105  5.441822e-01
floors       4.322975e+03  3.922318e+03   1.1021480  2.704096e-01
waterfront   5.700300e+05  1.949115e+04  29.2455818  1.994915e-184
view        5.402638e+04  2.364327e+03  22.8506357  3.254220e-114
condition    6.027619e+04  2.431933e+03  24.7853031  9.526113e-134
grade        1.125942e+05  2.346371e+03  47.9865299  0.000000e+00
sqft_above   -2.815739e+01  4.725441e+00  -5.9586797  2.582270e-09
yr_built     -4.159519e+02  8.788735e+00 -47.3278448  0.000000e+00
yr_renovated  6.706261e+01  3.871919e+00  17.3202511  9.352760e-67
sqft_living15 1.678526e+01  3.761002e+00  4.4629770  8.123729e-06
sqft_lot15   -7.131533e-01  8.187134e-02  -8.7106595  3.234424e-18
> summary(lm.result4)$r.squared
[1] 0.8803596
> summary(lm.result4)$adj.r.squared
[1] 0.8802821

```

6. The average sales price is 1689762, and the 95% predicted interval is [1235440, 2144084].



```

20 lm.result3 <- lm(price ~ bedrooms + bathrooms + I(sqft_living + waterfront + grade),
21 data = kc_house_data)
22 summary(lm.result3)$coefficients
23 summary(lm.result3)$r.squared
24 summary(lm.result3)$adj.r.squared
25
26 #5 Build a linear model to forecast price using all other columns and no y intercept
27 str(kc_house_data)
28 kc_house_data$date <- as.factor(kc_house_data$date)
29 lm.result4 <- lm(price ~ -1 + ., data = kc_house_data)
30 summary(lm.result4)$coefficients
31 summary(lm.result4)$r.squared
32 summary(lm.result4)$adj.r.squared
33
34 #6 Build a linear model to forecast price using bedrooms, bathrooms, sqft_living,
35 # sqft_lot, floors, waterfront, view, condition, and grade. Enter the new info:
36 # bedrooms = 4, bathrooms = 2, sqft_living = 2560, sqft_lot = 7650, floors = 1.5,
37 # waterfront = 1, view = 3, condition = 5, and grade = 10. Then find the average
38 # sales price and 95% predicted interval
39 lm.result5 <- lm(price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors
40 + waterfront + view + condition + grade, data = kc_house_data)
41 new.house <- data.frame(bedrooms = 4, bathrooms = 2, sqft_living = 2560,
42 sqft_lot=7650, floors = 1.5, waterfront = 1, view = 3, condition = 5, grade = 10)
43 predict(lm.result5, newdata = new.house, interval = "predict")
44
~/
> lm.result5 <- lm(price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors
+ + waterfront + view + condition + grade, data = kc_house_data)
> new.house <- data.frame(bedrooms = 4, bathrooms = 2, sqft_living = 2560,
+ sqft_lot=7650, floors = 1.5, waterfront = 1, view = 3, condition = 5, grade = 10)
> predict(lm.result5, newdata = new.house, interval = "predict")
      fit      lwr      upr
1 1689762 1235440 2144084
>

```